

THE TAMIL NADU Dr. M.G.R. MEDICAL UNIVERSITY
No. 69, ANNA SALAI, GUINDY, CHENNAI – 600 032.

B.D.S.
DEGREE COURSES



SYLLABUS AND CURRICULUM

THE TAMIL NADU Dr. M.G.R. MEDICAL UNIVERSITY, CHENNAI

PREFACE

The Syllabus and Curriculum for the B.D.S.Courses have been restructured with the Experts from the concerned specialities to educate students of BDS courseto

1. Take up the responsibilities of dental surgeon of first contact and be capable of functioning independently in both urban and rural environment.
2. Provide educational experience that allows hands-on-experience both in hospital as well as in community setting.
3. Make maximum efforts to encourage integrated teaching and de-emphasize compartmentalisation of disciplines so as to achieve horizontal and vertical integration in different phases.
4. Offer educational experience that emphasizes health rather than only disease.
5. Teach common problems of health and disease and to the national programmes.
6. Use learner oriented methods, which would encourage clarity of expression, independence of judgement, scientific habits, problem solving abilities, self initiated and self-directed learning.
7. Use of active methods of learning such as group discussions, seminars, role play, field visits, demonstrations, peer interactions etc., which would enable students to develop personality, communication skills and other qualities towards patient care.

The Students passing out of this Prestigious University should be acquire adequate knowledge, necessary skills and such attitudes which are required for carrying out all the activities appropriate to general dental practice involving the prevention, diagnosis and treatment of anomalies and diseases of the teeth, mouth, jaws and associated tissues. The students should also understand the concept of community oral health education and be able to participate in the rural health care delivery programmes existing in the country.

(Subject to changes in Amendments in DCI Regulations and SAB Resolutions)

**Prof. Dr.S.GEETHALAKSHMI, M.D., Ph.D.
VICE-CHANCELLOR**

Comments / Feed back are welcome if any and mail it to registrar@tnmgrmu.ac.in

B.D.S. - DEGREE COURSE

SECOND YEAR SUBJECTS

Sl. No.	Subjects	Page. No.
	II Year	
1.	General Pathology and Microbiology	1 - 12 13 - 21
2.	General and Dental Pharmacology and Therapeutics	22 - 27
3.	Dental Materials	28 - 46
4.	Pre Clinical Conservative Dentistry	47 - 54
5.	Pre Clinical Prosthodontics & Crown & Bridge	55 - 65

4. GENERAL PATHOLOGY

1. GOAL

At the end of the course the student should be competent to:

Apply the scientific study of disease processes, which result in morphological and functional alterations in cells, tissues and organs to the study of pathology and the practice of dentistry.

2. OBJECTIVES

a. KNOWLEDGE AND UNDERSTANDING:

- To demonstrate and analyze pathological changes at macroscopic and microscopic levels and explain their observations in terms of disease processes.
- To integrate knowledge from the basic sciences, clinical medicine and dentistry in the study of Pathology.
- To demonstrate understanding of the capabilities and limitations of morphological pathology in its contribution to medicine, dentistry and biological research.
- To demonstrate ability to consult resource materials outside lectures, laboratory and tutorial classes.

b. SKILLS:

- A dental graduate should be able to identify the abnormal diseases like tumor, non tumours and also to arrive what are the investigations needed for the diagnosis of the diseases.
- Carry out certain investigations and ability to interpret lab findings.

c. ATTITUDE:

- A dental student must be willing to apply the knowledge gained in pathology in the best interest of the patient and the community.
- Maintain a high standard of professional ethics In patient care and also in carrying out the diagnostic modalities.
- Willing to update knowledge in pathological conditions and diagnostic investigations from time to time.

d. INTEGRATION

The dental student must be able to integrate the pathological aspects with the diseases so that it helps to understand the disease nature and management of the disease.

e. COMPUTER PROFICIENCY

Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes. Basic operative skills in analysis of data and knowledge of multimedia. Students should utilize a combination of traditional classroom courses and online courses. The following validation is required and must be completed.

- i. Technological Requirements for all Graduate Students
- ii. A laptop or desktop computer that supports the following requirements
 - a. Operating system requirements
 - b. Internet browser requirements
 - c. Reliable and consistent access to the internet
 - d. Antivirus software which is current and consistently updated
 - e. Microsoft Office
 - f. Adobe Reader (or equivalent to view PDF files)

f. KNOWLEDGE ABOUT INFECTION AND CROSS INFECTION IN DENTISTRY

Knowledge about asepsis – disinfection and sterilization of instruments, clinical area / personal care as per universal protection, and disposal of medical wastes in the appropriate modes. Students should be aware of the rules and regulations pertaining to maintenance of clinical set up and waste disposal.

3. **COMPETENCIES**

1. General skills
2. Practice Management
3. Communication and Community Resources
4. Patient Care – Diagnosis
5. Patient Care - Treatment Planning
6. Competencies specific to subject

4. TEACHING HOURS

Lecture hours - **55**

Practical hours - **55**

Total hours **110 hours**

5. TEACHING METHODOLOGY

Lectures, symposiums, vertical and horizontal integrated teachings, viva voce, CMEs etc. The objectives of teaching General Pathology can be achieved by various teaching techniques such as :

- a) Lectures
- b) Lecture Demonstrations
- c) Practical exercises
- d) Audio visual aids
- e) Small group discussions with regular feedback from the students
- f) Integrated Teaching
- g) Symposium and continuing medical education programmes

6. THEORY SYLLABUS

TOPIC	MUST KNOW	DESIRABLE TO KNOW	NICE TO KNOW
Introduction	Cellular responses to stress & noxious stimuli, cellular adaptation of growth & differentiation (hyperplasia, hypertrophy, atrophy & metaplasia) Cell injury and cell death (cause & mechanism of reversible & irreversible injury) Morphology of cell injury (reversible & necrosis), examples of cell injury and necrosis (ischemic, hypoxic, reperfusion and chemical injuries)	Historical aspects; definition of terms; introduction to pathology, its applications and role in patient management.	

	Apoptosis and sub-cellular responses to injury Intracellular accumulation, calcification & cellular aging; (Lipid, protein, glycogen and pigment accumulation; pathologic calcification; ageing)		
Inflammation/ Repair	Introduction to body's immune response (innate & adaptive immunity; cells and tissues of immune system; cytokines; structure & function of HLA) General features of inflammation; history; stimuli for acute inflammation; vascular events; cellular events - leucocyte adhesion and transmigration Continuation of cellular events (chemotaxis, phagocytosis, defects of leucocyte function); termination of acute inflammatory response; outcome of acute inflammation; morphological patterns of acute inflammation; Chemical mediators (vasoactive amines; plasma proteins; AA metabolites; PAF; cytokines; chemokines; leucotrienes; NO; free radicals & neuropeptides) Chronic inflammation (cause, morphological features; cells of chronic inflammation; granuloma; systemic effects of inflammation; consequences of excessive/defective inflammation) Repair (healing; scar formation; cutaneous wound healing); Repair (continued) (healing at special sites; factors		

	affecting wound healing)		
Haemodynamic disturbances	Oedema, Hypotension, congestion, haemorrhage & haemostasis Thrombosis & embolism Infarction, Shock		
Disorders of Immunity	Disorders of immunity – mechanisms of hypersensitivity, Graft Rejection Autoimmunity – SLE Primary & secondary immunodeficiency Amyloidosis	Rheumatoid arthritis, systemic sclerosis, Sjogren's, MCD,	
Neoplasia	Definition, nomenclature, biology of tumour growth, differences between benign & malignant tumours Tumour spread & epidemiology Molecular basis of Neoplasia (essential alterations for malignant transformation, oncogenes, suppressor genes) Evasion of apoptosis; defects in DNA repair, telomerase and angiogenesis; invasion & metastasis; dysregulation of genes) Carcinogenesis (carcinogenic agents, molecular basis of carcinogenesis) Host defense, tumour immunity, clinical features, and laboratory diagnosis.		
Infectious	Mycobacterial infections – tuberculosis HIV & Hepatitis	Typhoid, syphilis	General principles

diseases	Viruses	and others Fungal & parasitic infections	(categories, transmission & dissemination of microbes, mechanisms of microbial disease, immune evasion, infections in immunosuppressed hosts, tissue response to microbes) Pathology of common viral & bacterial infections (CMV, EBV, HPV, viruses, gram positive & negative bacterial infections)
Nutritional		Nutritional diseases	
RBC & bleeding disorders	Development of haematopoietic cells, bone marrow, classification of anaemia Iron deficiency anaemia, Megaloblastic anaemia Bleeding disorders – classification, disorders of platelets Coagulation disorders		
WBC, lymph node, spleen	Leukaemia – classification, aetiology, acute leukaemias. Chronic leukaemias, MDS, other chronic myelo-	Non-neoplastic quantitative and qualitative disorders of	

	proliferative disorders including myelofibrosis Hodgkin Lymphoma Blood banking	leucocytes Non-neoplastic disorders of lymph node, spleen & thymus; classification of lymphoma	
Systemic Pathology	Atherosclerosis Hypertension, vasculitis	Congenital anomalies, aneurysms, tumors.	
The Heart	Ischemic heart disease & myocardial infarction Rheumatic fever; Infective endocarditis	Congenital heart disease, diseases of the myocardium, tumors of the heart; diseases of the pericardium	
Head and neck	Benign and malignant lesions of head and neck including oral cavity, salivary glands		
Kidney	Nephrotic syndrome – pathogenesis and pathology	Normal structure, congenital anomalies, cystic disease, laboratory tests in renal disease.	
Endocrine system	Diabetes mellitus		
Bone & Joints	Infections, metabolic disease of bone Bone tumours/Jaw tumours		

Bioethics

Bioethics is the application of ethics to the field of medicine and healthcare. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; Environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment and public health ethics.

7. PRACTICALS:

PROCEDURES:

1. Urine – Tests for Abnormal constituents Sugar, albumin, ketone bodies, Blood, bile salts, bile pigments.
2. Haemoglobin (Hb) estimation as OSPE
3. Total WBC count from the peripheral smear
4. Differential WBC Count and commenting on the peripheral smear
5. Blood grouping as OSPE

DEMONSTRATIONS

6. Packed cell volume(PCV,) Erythrocyte Sedimentation Rate (ESR)
7. Bleeding Time & Clotting Time
8. Histopathology Tissue Processing Staining
9. Histopathology slides
 - Acute appendicitis
 - Granulation tissue
 - fatty liver
 - CVC lung
 - CVC liver
 - CVC Spleen
 - Lipoma
 - Teratoma
 - Tuberculosis of Lymph node
 - Maduramycosis
 - Actionomycosis

Rhinosporidiosis
Basal cell Carcinoma
Squamous cell Carcinoma
Malignant melanoma,
Ameloblastoma,
Squamous papilloma
Hodgkins Lymphoma
Pleomorphic adenoma
Cavernous hemangioma
Capillary hemangioma
Osteosarcoma
osteoclastoma

HEMATOLOGY SLIDES

Iron deficiency anemia
Acute Myeloid Leukemia
Chronic Myeloid Leukemia
Eosinophila

LIST OF SPECIMENS:

- i. acute appendicitis
- ii. Fatty liver
- iii. CVC lung
- iv. CVC Liver
- v. Infarct spleen
- vi. TB lymph Node
- vii. Lipoma
- viii. Myxoma
- ix. Chondroma
- x. Squamous cell carcinoma
- xi. Pleomorphic adenoma

- xii. Teratoma
- xiii. Malignant Melanoma

Instruments:

- i. RBC Pipette
- ii. WBC Pipette
- iii. ESR Westergrens tube
- iv. SAHLI'S hemoglobinometer
- v. PCV tube
- vi. Bone marrow biopsy needle
- vii. Bone marrow aspiration needle

8. THEORY EXAMINATION (TITLE AND QP PATTERN WITH MARKS)

Part A - Pathology:

Essay	1X10 = 10 Marks
Short notes	3X 5 = 15 Marks
Short Answers	5X2 = 10 Marks

Total = 35 Marks

9. PRACTICAL EXAMINATIONS- experiments, slides and OSPE

Lab experiments 45 marks

Major experiment – Hematology -

Peripheral smear/ DC - 15 Marks, 45 Minutes

Urine analysis - 10 Marks, 30 Minutes

Minor experiment(OSPE) - 10 Marks, 20 Minutes (for Hb%)

Spotters - 10 Marks, 20 minutes

Total 45 marks

Viva - 10 marks

SPOTTERS:

- i. Histo pathology slides
- ii. Haematology slides
- iii. Gross specimens
- iv. Instruments

Scheme for practical examinations

Procedure

Demonstrations

Viva

	Examination	Internal Assessment	Viva	Total
Theory	35	5	10	50
Practicals	45	5	-	50
Total				100

10. FORMATIVE/INTERNAL ASSESSMENT

The continuing assessment examination (both Theory/Practical) held at least 3times in a particular year and best of two examinations shall be considered. The Internal Assessment marks to be submitted to the University, once in every three months. The marks scored by the students shall be displayed on the Notice board and a copy forwarded by HOD shall be sent to the University once in every 3 months.

Topics:

- i. Cell injury and adaptations,Inflammation wound healing
- ii. Hemodynamic changesNeoplasia
- iii. Infectious diseasesNutritional disorders

- iv. Disorders of circulations, Immunity, Diseases of oral cavity
- v. Diseases of the salivary glands, Bones, cardiovascular system
- vi. Hematology(RBC, WBC AND PLATELETS, LYMPHNODE, SPLEEN AND THYMUS)

Theory - 5 Marks

Practical - 5 Marks

Total - 10 marks

11. RECORD NOTE / LOG BOOK:

Record shall be maintained and assessed periodically by faculty and HOD. Institution shall provide adequate number of cases/teaching materials as specified in Dental Council of India regulation for the students during clinical/practical training and examinations.

12. TEXT BOOKS

- i. Robbins BASIC PATHOLOGY – by Kumar, Abbas and Aster- 1st South Asia edition
- ii. Text book of Pathology By Harsh Mohan 7th Edition
- iii. Andersons pathology Volume 1 And 2 by Ivan Damjanov & James Linder
- iv. 3.Wintrobe's Clinical Hematology by Lee, Bithell,Forster.

13. REFERENCE BOOKS:

- i. Robbins – Pathologic Basis of Diseases By Kumar and Kotran 10th Edition.
- ii. Ackermann Surgical Pathology
- iii. Microbiology – Prescott, et al.
- iv. Microbiology – Bernard D. Davis, et al.
- v. Clinical & Pathogenic Microbiology – Barbara J Howard, et al.
- vi. Mechanisms of Microbial diseases – Moselio Schechter, et al.
- vii. Immunology an Introduction – Tizard
- viii. Immunology 3rd edition – Evan Roitt, et al.

MICROBIOLOGY

1. GOAL

To introduce the students to the exciting world of microbes and to provide an understanding of various branches of Microbiology, in order to deal with the etiology, pathogenesis, laboratory diagnosis, treatment, control and prevention of infections in dental practice.

2. OBJECTIVES

a. KNOWLEDGE AND UNDERSTANDING:

At the end of the Microbiology course the student is expected to

- i. Understand the basics of various branches of Microbiology and able to apply the knowledge relevantly.
- ii. Apply the knowledge gained in related medical subjects like General Medicine and General Surgery and Dental subjects like Oral Pathology, Community Dentistry, Periodontics, Oral Surgery, Pedodontics, Conservative Dentistry and Oral Medicine in higher classes.
- iii. Understand and practice various methods of Sterilisation and disinfection in dental clinics.
- iv. Have a sound understanding of various infectious diseases and lesions in the oral cavity.
- v. Awareness of Health care associated infections and their prevention in dental practice

b. SKILLS

- i. Student should have acquired the skill to diagnose, differentiate various oral lesions.
- ii. Should be able to select, collect and transport clinical specimens to the laboratory.
- iii. Should be able to carry out proper aseptic procedures in the dental clinic.
- iv. Interpretation of antimicrobial susceptibility tests and to make right choice of antibiotic based on spectrum of infection and ensure appropriate use to avoid antibiotic resistance.

c. ATTITUDE:

- i. To apply knowledge in the interest of the individual patient and community.
- ii. Maintain high standards of professional ethics in patient care and in carrying out diagnostic tests.

iii. To update knowledge from time to time with regard to diagnostics and immunoprophylaxis.

d. INTEGRATION:

At the end of integrated teaching the student shall acquire integrated knowledge from different disciplines which includes etiology, morphology, pathogenesis, clinical features, laboratory diagnosis, treatment, prevention and control of infectious diseases.

e. KNOWLEDGE ABOUT INFECTION AND CROSS INFECTION IN DENTISTRY

Knowledge about asepsis – disinfection and sterilisation : of instruments , clinical area/ personal care as per universal protection, and disposal of medical wastes in the appropriate modes. Students should be aware of the rules and regulations pertaining to maintenance of clinical set up and waste disposal.

f. COMPUTER PROFICIENCY:

Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes Basic operative skills in analysis of data and knowledge of multimedia. Students should utilize a combination of traditional classroom courses and online courses. The following validation is required and must be completed.

- i. Technological Requirements for all Graduate Students
- ii. A laptop or desktop computer that supports the following requirements
 - a) Operating system requirements
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 - d) Antivirus software which is current and consistently updated
 - e) Microsoft Office
 - f) Adobe Reader (or equivalent to view PDF files)

3. COMPETENCIES

1. General skills
2. Practice Management

3. Communication to Community Resources
4. Patient Care – Diagnosis
5. Patient Care - Treatment Planning
6. Competencies specific to the subject

4. TEACHING HOURS

- Lecture hours 65
- Practical hours 50
- Total hours 115

5. TEACHING METHODOLOGY

The objectives of teaching microbiology can be achieved by various teaching techniques such as :

- a) Lectures
- b) Lecture Demonstrations
- c) Practical exercises
- d) Audio visual aids
- e) Small group discussions with regular feed back from the students
- f) Integrated Teaching
- g) Symposium and continuing medical education programmes.

6. THEORY SYLLABUS

TOPIC	MUST KNOW	DESIRABLE TO KNOW	NICE TO KNOW
Introduction, History	Noble laureates and their contributions to medical microbiology, Detailed contributions of Louis Pasteur, and Robert Koch		
	Morphology physiology, classification of bacteria, different methods of staining		
	Sterilization and disinfection including sterilization controls		

	Different types of culture media and culture techniques including anaerobic culture methods.	Bacterial genetics and drug resistance in bacteria	
	Specimen Collection, Transport processing and Identification of bacteria		Testing of disinfectants
	Infection-source, mode of transmission and types of infectious disease		
Immunology	1.Immunity 2.Antigen 3.Immunoglobulins 4.Structure and functions of immune system 5.Antigen -Antibody reactions 6.Immune response 7.Hypersensitivity 8. Auto immunity, classification with special reference to autoimmune disorders involving oral cavity. 9.Immunodeficiency disorders-various types and disorders relevant to dentistry 10.Immunology of transplantation and malignancy	Complement system Immunohaematology	Flow cytometry in the diagnosis of malignancies Vaccines against tumors
Systematic bacteriology	1.Gram positive cocci - Staphylococcus, Streptococcus with special reference to Viridans group, Pneumococcus 2.Gram negative cocci – Meningococcus and Gonococcus 3.Corynebacterium diphtheria including immunoprophylaxis 4.Clostridium – Gas Gangrene, Tetanus and food poisoning 5.Mycobacteria- M.tuberculosis and M.leprae 6. Non sporing anaerobes – classification , pathogenesis, Laboratory diagnosis and treatment.	Enterobacteriaceae Vibrio cholera	MDR and XDR TB Agents of Bioterrorism

	7.Spirochaetes- Treponema, Borrelia vincenti 8.Actinomycetes 9.Normal flora of oral cavity		
Virology	1.General properties, resistance cultivation of viruses, host virus interactions with special reference to interferon 2.Laboratory diagnosis , Viral vaccines 3.Herpes virus 4.Measles , Mumps and Rubella 5.Rabies virus 6.Hepatitis B and Hepatitis C virus,HBV vaccine 7.Human Immunodeficiency virus	Bacteriophage structure and significance Cultivation of viruses	Influenza A and B viruses
Mycology	1.Introduction,classification, Laboratory diagnosis 2.Candidosis,Rhinosporidiosis 3.Systemic mycoses and associated oral lesions.	Opportunistic fungal infections	Antifungal susceptibility testing methods
Parasitology	1.Introduction , different modes of transmission and prevention 2.Entamoeba histolytica, Entamoeba gingivalis 3.Malarial parasites 4.Leishmania including L.brasiliensis 5.Common helminthic infections – Tape worms, Ascaris lumbricoides, Ancylostoma duodenale, Trichuris trichura and Enterobius vermicularis.	Protozoa Giardia intestinalis, Trichomonas species. Wuchereria bancrofti	Parasitic infections in HIV
Applied Microbiology	1.Standard precautions 2.Infection control measures in dental setting 3.Significance of antibiotic susceptibility testing ,its interpretation 4.Bio medical waste management guidelines 5..Vaccination for Health care providers 6..Needle stick injury and post exposure prophylaxis 7.Blood borne infections	STD infections Infective endocarditis Emerging and Re emerging infections	Antibiotic resistance (MRSA,ESBL etc.)

Bioethics

Bioethics is the application of ethics to the field of medicine and healthcare. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment, and public health ethics.

In microbiology, the maintenance of confidentiality is very important for the laboratory to gain confidence from the patients. Confidentiality is mandatory in certain tests like HIV testing as the results may lead to alienation from the family thus causing mental agony to the patient. Counselling has to be given both before and after testing in HIV /AIDS setting. Written consent has to be always obtained from the patient for any procedure that can potentially harm the individual particularly invasive techniques.

Quarantining of people is done under special circumstances. By adhering to ethical guidelines, members of the medical profession can help and ensure that quarantine and isolation measures achieve their public health goals and maximally promote the well-being of individuals.

7. PRACTICALS

Procedures

- i. Simple stain, Hanging drop
- ii. Grams stain
- iii. Ziehl Neilsen's stain

Demonstrations

- i. Microscopy-Different types, parts, maintenance and usage
- ii. Sterilization and disinfection
- iii. Culture media including anaerobic culture media and transport media
- iv. Anaerobic culture methods
- v. Biochemical reactions in the identification of bacteria
- vi. Virus models

8. THEORY EXAMINATION

Part B – Microbiology:

Essay	1 X 10	=	10 Marks
Short Notes	3 X 5	=	15 Marks
Short Answers	5 X 2	=	10 Marks

Total		=	35 Marks

Note: Essay from Systematic Bacteriology/Virology, General bacteriology Immunology
Short Notes from Systematic bacteriology, Virology, Mycology, Parasitology, Applied Microbiology
Short Answers from General bacteriology, Immunology, Systematic bacteriology, Virology, Mycology, Parasitology and Applied Microbiology.

9. PRACTICAL EXAMINATION

Contents	Marks	Time duration
Spotters (10x 2marks each)	20	30mts
Gram staining (GPC,GNB,MIXTURE)	10	45 mts
Ziehl Neilsen's staining	10	60mts
*OSPE	5	45mts
Total	45marks	180mts(3hrs)

*OSPE Exercises Eg. Hand washing Technique

Bio medical waste segregation

OR any other relevant topic of choice

Note : For OSPE,key to be prepared and made available to the examiners .

Viva – Marks 10

To be conducted in the afternoon with appropriate time interval.

	Examination	Internal Assessment	Viva	Total
Theory	35	5	10	50
Practicals	45	5	-	50
Total				100

10. FORMATIVE /INTERNAL ASSESSMENT

The continuing assessment examination (both Theory/Practical) held at least 3times in a particular year and best of two examinations shall be considered. The Internal Assessment marks to be submitted to the university, once in every three months. The marks scored by the students shall be displayed on the Notice board and a copy forwarded by HOD shall be sent to the University once in every 3 months.

Theory - 5 marks

Practicals - 5 marks

Total - 10 marks

11. RECORD NOTE / LOG BOOK

Record shall be maintained and assessed periodically by faculty and HOD. Institution shall provide adequate number of cases/teaching materials as specified in Dental Council of India regulation for the students during clinical/practical training and examinations.

12. TEXT BOOKS

- i. Text book of Microbiology – R.Ananthanarayan & C.K.Jayaram Paniker.
- ii Medical Microbiology – David Greenwood etal.
- iii. Textbook of parasitology – K.D.Chatterjee
- iv. Paniker's Text book of Medical Parasitology

13.BOOKS FOR FURTHER READING/REFERENCE.

- i. Microbiology – Prescott, etal.
- ii. Microbiology – Bernard D. Davis , etal.
- iii. Clinical & Pathogenic Microbiology – Barbara J Howard, etal.

- iv. Mechanisms of Microbial diseases – Moselio Schaechter, etal.
- v. Immunology –Donald M Weir
- vi. Immunology 3rd edition – Evan Roitt , etal.
- vii. Oral microbiology and infectious diseases –Burnett and Scherp
- viii. Jawetz text book of microbiology

5. GENERAL AND DENTAL PHARMACOLOGY AND THERAPEUTICS

1. GOAL

The broad goal of teaching undergraduate students in pharmacology is to inculcate rational and scientific basis of therapeutics keeping in view of dental curriculum and profession.

2. OBJECTIVES

a) KNOWLEDGE AND UNDERSTANDING:

At the end of the course the student shall be able to

- i. Describe the pharmacokinetics and pharmacodynamics of essential and commonly used drugs in general and in dentistry in particular.
- ii. List the indications, contraindications, interactions and adverse reactions of commonly used drugs with reason.
- iii. Tailor the use of appropriate drugs in disease with consideration to its cost, efficacy, safety for individual and mass therapy needs.
- iv. Indicate special care in prescribing common and essential drugs in special medical situations such as pregnancy, lactation, old age, renal, hepatic damage and immunocompromised patients.
- v. Integrate the rational drug therapy in clinical pharmacology.
- vi. Indicate the principles underlying the concepts of “Essential drugs”.

b) SKILLS:

At the end of the course student shall be able to:

- i. Prescribe drugs for common medical and dental ailments.
- ii. Appreciate adverse reactions and drug interactions of commonly used drugs
- iii. Observe experiments designed for study of effects of drugs.
- iv. Critically evaluate drug formulations and be able to interpret the clinical pharmacology of marketed preparations commonly used in dentistry.

c) ATTITUDE:

To develop the attitude to serve the rural community

d) INTEGRATION:

Practical knowledge of use of drugs in clinical practice will be acquired through integrated teaching with clinical departments

e) KNOWLEDGE ABOUT INFECTION AND CROSS INFECTION IN DENTISTRY

Knowledge about asepsis – disinfection and sterilization of instruments, clinical area / personal care as per universal protection and disposal of medical wastes in the appropriate modes. Students should be aware of the rules and regulations pertaining to maintenance of clinical set up and waste disposal.

f) COMPUTER PROFICIENCY

Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes. Basic operative skills in analysis of data and knowledge of multimedia. Students should utilize a combination of traditional classroom courses, and online courses. The following validation is required and must be completed.

- i. Technological Requirements for all Graduate Students
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 - d) Antivirus software which is current and consistently updated
 - e) Microsoft Office
 - f) Adobe Reader (or equivalent to view PDF files)

3. COMPETENCIES

1. General skills
2. Practice Management
3. Communication and Community Resources
4. Patient Care – Diagnosis
5. Patient Care - Treatment Planning
6. Competencies Specific to the subject

4. TEACHING HOURS

Lecture hours - 70 hours

Practical hours- 20 hours

Total – 90 hours

5. TEACHING METHODOLOGY

The objectives of teaching can be achieved by various teaching techniques such as :

- a) Lectures
- b) Lecture Demonstrations
- c) Practical exercises
- d) Audio visual aids
- e) Small group discussions with regular feed back from the students
- f) Integrated Teaching
- g) Symposium and continuing medical education programmes.

6. THEORY SYLLABUS

- New drug development- clinical trials, biomedical ethics;
- Pharmacoeconomics;
- Pharmacovigilance

SYSTEMIC PHARMACOLOGY

TOPIC	MUST KNOW	DESIRABLE TO KNOW	NICE TO KNOW
1.	GENERAL PHARMACOLOGY	DRUGS ACTING ON CARDIOVASCULAR SYSTEM	VITAMINS: Water soluble vitamins, vitamin D, vitamin K, vitamin E, implications of vitamins in clinical dentistry.
2.	ANTIBIOTICS	DRUGS ACTING ON CENTRAL NERVOUS SYSTEM	VACCINES
3.	NSAIDS	DIURETICS	
4.	DRUGS ACTING ON GI TRACT	DRUGS ACTING ON BLOOD	
5.	LOCAL ANESTHETICS	GENERAL ANESTHETICS	
6.	DRUGS ACTING ON AUTONOMIC NERVOUS SYSTEM	ANTINEOPLASTIC AGENTS	
7.	INSULIN AND ORAL HYPOGLYCAEMIC DRUGS		
8.	CORTICOSTEROIDS		
9.	ANTISEPTICS AND DISINFECTANTS		

Bioethics

Bioethics is the application of ethics to the field of medicine and healthcare. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment, and public health ethics.

7. PRACTICALS

Procedures and demonstrations:

To familiarize the student with prescription writing and dispensing. Rational of drug combinations of marketed drugs

8. THEORY EXAMINATION

Elaborate on 2x10= 20 marks
Write notes 10x5 = 50 marks
Total = 70 marks

9. PRACTICAL EXAMINATION

Dispensing pharmacy 2x25= 50 marks
Prescription 2x10= 20 marks
OSPE 2x 10=20 marks
Total 90 marks

Viva 20 marks

	Examination	Internal Assessment	Viva	Total
Theory	70	10	20	100
Practicals	90	10	-	100
Total				200

10. FORMATIVE / INTERNAL ASSESSMENT

The continuing assessment examination (both Theory/Practical) held at least 3times in a particular year and best of two examinations shall be considered. The Internal Assessment marks to be submitted to the university, once in every three months. The marks scored by the students shall be displayed on the Notice board and a copy forwarded by HOD shall be sent to the University once in every 3 months.

Theory 10 marks
Practicals 10 marks
Total 20 marks

Topics for Internal Assessment

- i. General Pharmacology
- ii. Autonomic Nervous system
- iii. Central Nervous system
- iv. Cardiovascular system
- v. Respiratory system, Gastrointestinal system, autotoxins
- vi. Hormones
- vii. Chemotherapy

11. RECORD NOTE / LOG BOOK

Record shall be maintained and assessed periodically by faculty and HOD. Institution shall provide adequate number of cases/ teaching materials as specified in Dental Council of India regulation for the students during clinical /practical training and examinations.

12. TEXT BOOKS

- i. Tripathi K D – Essentials of medical pharmacology
- ii. R S Satoskar- Pharmacology and Pharmacotherapeutics
- iii. Bertam G Katzung- Basic and clinical pharmacology

13. REFERENCE BOOKS

- i. Goodman and Gilman- The Pharmacological basis of Therapeutics.
- ii. R.S.Satoskar, Kale Bhandarkar's Pharmacology and Pharmacotherapeutics, 10th Edition, Bombay Popular Prakashan 1991.
- iii. Bertam G Katzung, basic and Clinical pharmacology 6th ed.Appleton & Lange 1997.
- iv. Lauerence D.R. Clinical Pharmacology 8th ed. Churchill Livingstone 1997.
- v. Satoskar R.S. & Bhandarkar S.D., Pharmacology and Pharmacotherapeutics part I & part ii, 13th Popular Prakashan Bombay 1993.
- vi. Tripathi K.D., Essentials of Medical Pharmacology 4th ed Jaypee Brothers 1999.

6. DENTAL MATERIAL

1. GOAL

The dental graduates during training in the institutions should acquire adequate knowledge, necessary skills and such attitudes which are required for carrying out all the activities appropriate to general dental practice involving the prevention, diagnosis and treatment of anomalies and diseases of the teeth, mouth, jaws and associated tissues. Aim of the course is to present basic chemical and physical properties of dental materials as they are related to its manipulation to give a sound educational background about the various materials. The broad goal of the teaching of undergraduate students in Dental Materials aims at providing adequate fundamental knowledge about the materials available in the Dental science. .

2. OBJECTIVES

The objectives are dealt under three headings namely (a) knowledge and understanding (b) skills and (c) attitudes.

a. KNOWLEDGE AND UNDERSTANDING:

The graduate should acquire the following during the period of training --- Adequate knowledge of the scientific foundations on which dentistry is based and good understanding of various relevant scientific methods, principles of biological functions and should be able to evaluate and analyse scientifically various established facts and data. To understand the evolution and development of science of dental materials. To know about the manipulation technique of various restorative materials.

b. SKILLS:

A graduate should be able to demonstrate the following skills necessary for practice of dentistry. To develop skills in the management of various materials in dentistry. Students should know about the physical and chemical properties of the dental materials

c. ATTITUDE:

A graduate should develop during the training period the following attitudes. Willing to apply current knowledge of dentistry in the best interest of the patients and the community. Maintain a high standard of professional ethics and

conduct and apply these in all aspects of professional life. Seek to improve awareness and provide possible solutions for oral health problems and needs throughout the community. Willingness to participate in the continuing education programmes to update knowledge and professional skills from time to time. To help and to participate in the implementation of National Health Programmes.

d. INTEGRATION:

e. KNOWLEDGE ABOUT INFECTION AND CROSS INFECTION IN DENTISTRY:

Knowledge about asepsis – disinfection and sterilization of instruments, clinical area / personal care as per universal protection, and disposal of medical wastes in the appropriate modes. Students should be aware of the rules and regulations pertaining to maintenance of clinical set up and waste disposal.

f. Computer Proficiency

Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes. Basic operative skills in analysis of data and knowledge of multimedia. Students should utilize a combination of traditional classroom courses, and online courses. The following validation is required and must be completed

- i. Technological Requirements for all Graduate Students
- ii. A laptop or desktop computer that supports the following requirements
 - a) Operating system requirements
 - b) Internet browser requirements
 - c) Reliable and consistent access to the internet
 - d) Antivirus software which is current and consistently updated
 - e) Microsoft Office
 - f) Adobe Reader (or equivalent to view PDF files)

3. COMPETENCIES

1. General skills
2. Practice Management
3. Communication and Community Resources
4. Patient Care – Diagnosis

5. Patient Care - Treatment Planning
6. Competencies specific to the subject

4. TEACHING HOURS

Teaching hours for first and second years- Theory and Practical are shown in the Tables-I
TABLE - I Subjects and Hours of Instruction (B.D.S Course)

TOTAL TEACHING HOURS FOR FIRST AND SECOND B.D.S

Sl No	Subject	Lecture Hours	Practical Hours	Clinical Hours	Total HOURS
1.	Dental Materials	80	240	-	320

Subjects and Hours of Instruction for First year B.D.S

Sl No	Subject	Teaching Hours	Practical Hours	Clinical Hours	Total
1.	Dental Materials	20	40	—	60

Subjects and Hours of Instruction for Second year B.D.S

Sl No	Subject	Lecture Hours	Practical Hours	Clinical Hours	Total Hours
1.	Dental Materials	60	200	--	260

5. TEACHING METHODOLOGY

The objective of teaching can be achieved by various teaching tech such as

- i. Lecture
- ii. Demonstration
- iii. Practical exercises
- iv. Audio Video aids
- v. Group discussion
- vi. Integrated teaching

Titles of subjects of study

First Year

Dental Materials.

Second Year

Dental Materials.

6. THEORY SYLLABUS

TOPICS	MUST KNOW	DESIRABLE TO KNOW	NICE TO KNOW
Introduction	Brief History of the development of the science of Dental Materials. Aim of studying the subject of Dental Materials. Scope and requirements of Dental materials . Spectrum of materials - Classification Clinical and laboratory applications		
Structure of matter, and principles of adhesion	Change of state, inter atomic primary bonds, inter atomic secondary bonds, inter atomic bond distance and bonding energy, thermal energy, crystalline structure, ,non crystalline structures, diffusion, adhesion and bonding and adhesion to tooth structures.,	Change of state Interatomic bonds Crystalline structure Non crystalline solids and their structure	
Important Physical properties	Hue, value, chrome. and translucency physical properties based on laws of optics, dealing with phenomena of light, vision and sight. Thermal		

applicable to dental. Materials	conductivity & coefficient of thermal expansion, physical properties based on 'laws of thermodynamics. Stress, strain, proportional limit, elastic limit yield strength, modulus of elasticity, flexibility, resilience, impact, impact strength, permanent deformation, strength, flexure strength fatigue, static fatigue, toughness, brittleness, ductility & malleability, hardness, abrasion resistance, relaxation, rheology, Thixotropic, creep, static creep, dynamic 6reep, flow, colour, three dimensional colour - hue, values, chrome., Munsell system, metamerisim, fluorescence.		
Biological considerations in use of dental materials.	Classification of materials from perspective of biological compatibility	Micro leakage, Thermal changes, Galvanism, toxic effect of materials	Biological evaluation for systemic toxicity, skin irritation, mutagenicity and carcinogenicity.
Gypsum & gypsum products	Gypsum - its origin, chemical formula. Dental plaster, Dental stone, Die stone, high strength, high expansion stone. Application and manufacturing procedure of each, macroscopic and microscopic structure of each. Commercial names. Chemistry of setting, setting reaction, theories of setting, gauging water, Microscopic structure of set material. Setting time: working time and	Recent methods or advanced methods.	Disinfection of dental materials for infection control. Any recent advancements in material and mixing devices.

Impression materials used in dentistry	Measurement of setting time and factors controlling setting time. Setting expansion, Hygroscopic setting expansion Factors affecting each Strength: wet strength, dry strength, factors affecting strength. ADA classification of gypsum products Description of impression plaster and dental investment Manipulation Disinfection : infection control, liquids, sprays, radiation Method of use of disinfectants Storage of material - shelf life Impression plaster, Impression compound, Zinc oxide eugenol impression paste & bite registration paste incl., non eugenol paste, Hydrocolloids, reversible and irreversible, Elastomeric impression materials. Polysulphide, Condensation silicones, Addition silicones, Polyether. Definition of impression ., Purpose of making impression, Ideal properties required and application of material, Classification as per ADA specification, general & individual impression material. Application and their uses in different disciplines, Type of impression trays required, Adhesion, to Tray, manipulation, instruments & equipment's required. Techniques of impression, storage	Visible light cure polyether urethane dimethacrylate, Historical background , development Of each impression material,	
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Synthetic resins used in dentistry.	of impression, Working time, setting time, flow, accuracy, strength, flexibility, tear strength, dimensional stability, compatibility with cast & die materials incl., electroplating, Biological properties: tissue reaction Shelf life & storage of material, Infection control - disinfection, Advantages and disadvantages of each material. Classification of resins, Dental resins. Requirements of dental resins, applications, polymerisation, polymerisation mechanism. Stages in addition polymerisation, inhibition of polymerisation, copolymerisation, molecular weight, crosslinking, plasticisers. Physical properties of polymers, polymer structures types of resins. ACRYLIC RESINS: Mode of polymerisation: Heat activated, Chemically activated, Light activated, Mode of supply, application, composition, polymerisation reaction of each. Physical properties of denture base resin. Composite RESTORATIVE RESIN: Mode of supply, Composition, Polymerisation mechanisms: Chemically activated, Light activated, Dual cure: Degree of conversion, Polymerisation Shrinkage Classification of Composites: Application, composition and properties of each. Biocompatibility, -- micro leakage, pulpal reaction, pulpal protection Manipulation of composites:	Historical background and, development of material. Miscellaneous resins & techniques: Repair resins, Relining and rebasing. Infection control in detail, Biological properties and allergic reactions. Measurement of bond strength and micro leakage Amalgam Bonding Pit and fissure sealants.	Short term and long-term soft-liners, temporary crown and bridge, resins, Resin impression trays, Tray materials, Resin teeth, materials in maxillofacial prosthesis, Denture cleansers. Composites of posterior teeth, Prosthodontics resins for veneering. Repair of composite. Extended application for composites: Resins for restoring eroded teeth, Pit and fissure sealing, Resin inlay system Indirect & direct, Core build up, Orthodontic applications.
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	Techniques of Insertion of Chemically activated, light, activated, dual cure Polymerisation, Finishing and polishing of restoration, Direct Bonding: Need for bonding, Acid' etch technique,, Enamel bonding, Dentin bonding agents. Mode of bonding, Bond strength, Sandwich technique its indication and procedure.	Restorative Resins Depth of cure Degree of conversion, Dual Cure resins	Restorative Resins Curing lamps Depth of cure Reduction of residual stresses
Metal and alloys	Structure and behaviour of metals, Classification of casting alloys: By function & description. Alloys for crown & bridge, metal ceramic & removable partial denture. Composition,, function, constituents and application. Dental Amalgam Composition,Manufacturing of alloy powder,Amalgamation,Dimensional Stability,StrengthCreep,Clinicalperformance,Proportioning,Trituration,Condensation,Carving and finishing, Dimensional Change, Mercury hygiene	Historical background, desirable properties of casting alloys Factors affecting success of amalgam Side effects of mercury Repair of amalgam restoration	An alternative to metal casting process. Cad-cam process for metal & ceramic inlays
Direct filling gold	Properties of pure gold Classification and forms of DFG Removal of surface impurities	History, Compaction Direct gold restoration	

Dental casting alloys	Classification of casting alloys: By function & description. Recent classification High noble (HN); Noble (N) and predominantly base metal (PB). Alloys for crown & bridge, metal ceramic & removable partial denture. Composition, function, constituents and application, each alloy both noble and base metal. Properties of alloys: Melting range, mechanical properties, hardness, and elongation, modulus of elasticity, tarnish and corrosion. Casting shrinkage and compensation of casting shrinkage. Biocompatibility – Handling hazards. & precautions of base metal alloys, casting investments used. Heat treatment : Softening & hardening heat treatment	Historical background, desirable properties of casting alloys.	Alternatives to cast metal technology: direct filling gold, amalgam, mercury free, Condensable intermetallic compound - an alternative to metal casting process. CAD-CAM process for metal & ceramic inlays - without need for impression of teeth or casting Procedure, pure titanium, most bio compatible. metal 'which are difficult to cast can be made into crowns with the aid of CAD- CAM technology . Another method of making copings - by copy milling (without casting Procedures
Dental waxes including inlay casting wax	Introduction and importance of waxes. Sources of natural waxes and their chemical nature. Classification of Waxes: Properties of Dental wax, Inlay wax. Mode of supply composition, Ideal requirements. Properties: melting range, thermal expansion, mechanical properties, flow & residual stresses, ductility. Dental Wax: Inlay wax: Mode: Classification & composition,		Manipulation of inlay wax: Instruments & equipment required. Impression wax for corrective impressions, Bite registration wax.

Dental casting investments.	Ideal requirements: Properties of inlay wax: Flow, thermal properties Wax distortion & its causes. Definition, requirements, classification Gypsum bonded - classification. Phosphate bonded, 'Silica bonded'. Mode of Supply:, Composition, application , setting mechanism, setting time & factors controlling it. Expansions :Setting expansion, Hygroscopic Setting expansion, & thermal expansion : Factors affecting. Properties: Strength, porosity, and fineness & storage. Technical considerations:		Casting procedure, Preparation of die, Wax pattern, spruing, investing, and control of shrinkage compensation, wax burnout, and heating the invested ring, casting. Casting machines, source of heat for melting the alloy. Defects in casting.
Soldering, brazing and welding	Need of joining dental appliances, temperature, and application. Mode of supply of solders, Composition and selection, Properties. Tarnish & corrosion resistance mechanical properties, microstructure of soldered joint Fluxes & Anti fluxes: Definition, Function, Types, commonly used fluxes & their selection Welding: Definition, application, requirements, and procedure.	Technique of Soldering & Brazing : free hand soldering and investment, steps and Procedure.	weld decay - causes and how to avoid it. Laser welding.
Wrought base metal alloys	Applications and different alloys used mainly for orthodontics purpose 1. Stainless steel 2. Cobalt chromium nickel 3. Nickel titanium 4. Beta titanium		Titanium alloys, application, composition, properties, welding, Corrosion resistance

Dental cements	Properties required for orthodontic wires, working range, springiness, stiffness, resilience, Formability, ductility, ease of joining, corrosion resistance, stability in oral environment, biocompatibility Stainless steels: Description, type, composition & properties of each type. Sensitisation & stabilisation, Mechanical properties - strength, tensile, yield strength, KHN. Braided & twisted wires their need ;Solders for stainless steel, Fluxes, Welding 1.Wrought cobalt chromium nickel alloys, composition, allocation, properties, heat treatment, Physical properties 2. Nickel - Titanium alloys, shape, memory & super elastic Application, classification (general and individual), setting mechanism, mode of supply, Properties, factors affecting setting, special emphasis on critical procedures of manipulation and protection of cement, mode of adhesion, biomechanism of caries inhibition. Agents for pulpal protection. Definition & Ideal requirements. Fluoride releasing cements Luting cements Agents for pulp protection Zinc Phosphate cement Zinc Polycarboxylate Cement Glass ionomer cement		Modifications and recent advances, Principles of cementation. Special emphasis on cavity liners and cement bases and luting agents.
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Dental ceramics	Resin Cements Zinc oxide eugenol cement Calcium Hydroxide General applications. Dental ceramics: properties definition, classification,application, mode of supply, manufacturing procedure, methods of strengthening.Properties of fused ceramic:. Strength and factors affecting, modulus of elasticity, surface hardness, wear resistance, thermal properties, specific gravity, chemical stability, aesthetic properties, biocompatibility, technical considerations. Metal Ceramics (PFM): Alloys - Types and composition of alloys. Ceramic - Type and Composition.	Historical background. Methods of strengthening. Metal Ceramics (PFM).Metal Ceramic Bond.Metal Ceramic Bond - Nature of bond. Bonding using electro deposition, foil copings, bonded platinum foil, swaged gold alloy foil coping. Technical considerations of porcelain and porcelain fused metal restorations.	Recent advances - all porcelain restorations, Manganese core, injection moulded, cast able ceramics, glass infiltrated alumina core ceramic (In ceram), ceramic veneers, inlays and on lays, and CAD - CAM ceramic.
Abrasion & polishing agents	Definition of abrasion and polishing. Need of abrasion and polishing. Types of abrasives: Finishing, polishing & cleaning. Types of abrasives: Diamond, Emery, aluminium oxides garnet, pumice, Kieselgurh, tripoli, rouge, tin oxide, chalk, chromic	Technical consideration - Material and procedure used for abrasion and polishing,	

Die and counter die materials	oxide, sand, carbides, diamond, zirconium silicate, Zinc oxide		
Mechanics of cutting	Desirable 'characteristics of an abrasive, Rate of abrasion, Size of particle, pressure, Grading of abrasive & polishing agents. Binder, Polishing materials & procedures		
Dental implants	Types - Gypsum products, Electroforming, Epoxy resin, Amalgam. Burs and points.		Evolution of dental implants, - types and materials.

Bioethics

Bioethics is the application of ethics to the field of medicine and healthcare. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment, and public health ethics

BIO-ETHICS

- 1) Respect human life with dignity
- 2) Refrain from supporting crimes against humanity
- 3) Treat the sick with compassion
- 4) Protect the privacy of the patient
- 5) Educate the public
- 6) Fight for socio economical changes
- 7) Teaching and mentoring those who follow us

7. PRACTICAL

Practical Exercises: 240 Hours

Demonstration of manipulation of all materials

Exercises to be done by each student:

- a. Manipulation of Gypsum- Materials and Alginate - identify setting time and working time and working time with reference to proportion, water temp, and spatulation time.
- b. Self-cure and heat cure acrylic resin manipulation and curing.
- c. Cements - manipulation and studying setting time and working time for luting, base & restoration. Zinc oxide eugenol, zinc phosphate, glass ionomer .
- d. Silver Amalgam - manipulation, trituration.

8. THEORY EXAMINATIONS: (3 Hours)

Elaborate on 2 X 10 = 20 marks

Write Notes 10 X 5 = 50 marks

Total 70 marks

Note : One Elaborate on Question from Conservative Dentistry topics and one Elaborate on Question from Prosthodontics topics

Write Notes : Four Questions from conservative and four questions from Prosthetic topics and two questions from Metallurgy and orthodontia.

II Exercise to be done by each FIRST B.D.S student:

- | | | |
|-------------------------------------|---|----------|
| a. Impression material Manipulation | - | 20 hours |
| b. Gypsum products | - | 20 hours |

9. PRACTICAL / CLINICAL EXAMINATIONS:-

I. i. Spotters: Identify and write the composition and two important uses:

ii. Spotters – 20 Nos. 20 X 2 = 40 marks

Time – 2 Minutes each

II. Exercise No.1

Any one exercise of the following 25 Marks

- i. Manipulation of Dental plaster and stone
- ii. Manipulation of alginate impression material
- iii. Manipulation of Zinc Oxide Eugenol impression paste
- iv. Manipulation of heat cure acrylic resin

III. Exercise No. 2

25 Marks

Manipulation of any one of the following Dental Cements.

- a. ZOE (Luting and Filling consistency)
- b. Zinc Phosphate Cement (Luting and Base consistency)
- c. Glass Ionomer Cement Type I/II (Luting/Filling consistency)
- d. Silver Amalgam Trituration

TIMING FOR MANIPULATION

2-5 Minutes may be allotted for each mixing exercises

Viva

20 Marks

	Examination	Internal Assessment	Viva	Total
Theory	70	10	20	100
Practicals	90	10	-	100
Total				200

10. FORMATIVE / INTERNAL ASSESSMENT:

The continuing assessment examination (both Theory/Practical) held at least 3 times in a particular year and best of two examinations shall be considered. The Internal Assessment marks to be submitted to the university, once in every three months. The marks scored by the students shall be displayed on the Notice board and a copy forwarded by HOD shall be sent to the University once in every 3 months.

11. RECORD NOTE / LOG BOOK:

Record shall be maintained and assessed periodically by faculty and HOD. Institution shall provide adequate number of cases/teaching materials as specified in Dental Council of India regulation for the students during clinical/practical training and examinations.

12. TEXT BOOKS

Name of the Book & Title	Author	Edn	Yr. of Publ.	Publ.'s Name Place of Publ.
Science of Dental Materials	Kennet. J. Anusavice	11th	2007	W.B. Sunder's Company, USA
Notes on Dental Materials	E.C. Combe	06th	1992	Churchill Livingstone, UK Oxford Blackwell Scientific pub.
Applied Dental Material	John. F. Mc. Cabe	07th	1992	London
Text Book of Dental Material	Craig. O. Brien	06th	1996	Mosby, USA
Restorative Dental	Craig.	11th	2002	Mosby, USA

LIST OF SPOTTERS

CONSERVATIVE SPOTTERS:

1. Amalgam Alloy Powder

2. Mercury
3. Amalgam Capsule
4. Acid Etchant
5. Dentin Bonding Agent
6. Cavity Varnish
7. Dentin Conditioner
8. Composite Resin
9. Zinc Oxide Eugenol Cement
10. Modified Zinc Oxide Eugenol Cement (Irm – Intermediate Restorative Material)
11. Zinc Phosphate Cement
12. Zinc Polycarboxylate Cement
13. Glass Ionomer Cement Type I
14. Glass Ionomer Cement Type II
15. Calcium Hydroxide
16. Inlay Wax
17. Base Metal Alloy Pellets
18. Casting Ring
19. Gypsum Bonded Investment
20. Phosphate Bonded Investment
21. Dental Bur
22. Wooden Wedges
23. Gutta Percha Points
24. Gutta Percha Sticks
25. Motor And Pestle
26. Glass Slab
27. Cement Spatula
28. Agate Spatula

Prosthodontics spotters

1. plaster of paris
2. die stone
3. dental stone
4. gypsum bonded investment

5. zinc oxide eugenol impression paste
6. rubber base materials
7. alginate
8. impression compound
9. low fusing compound
10. sticky wax
11. shellac base plate
12. modelling wax
13. heat cure resin
14. self cure resin
15. metal pellets
16. casting ring
17. stainless steel wire
18. acrylic trimmers
19. separating media
20. acrylic teeth set
21. cotton puff
22. wollen puff
23. metal ceramic bridge

Miscellaneous

1. Infection control
2. Artificial tooth material.
3. Separating media
4. Die spacers
5. Tray adhesives
6. Petroleum jelly
7. Articulating paper
8. Pressure indicating paste
9. Endodontic materials
10. Comparative studies between metallic and nonmetallic denture base Bioglass
11. Sprues

12. Setting expansion, hygroscopic expansion, thermal expansion
13. Dentifrices.

13. REFERENCE BOOK:

1. Phillips Sciences of Dental Materials – 10th edn. –Kenneth J. Anusavice
2. Restorative Dental Material – 10 edn. Robert G.Craig
3. Notes on Dental Materials – E.C.Combe

7. PRE CLINICAL CONSERVATIVE DENTISTRY

1. GOAL

The IInd year BDS undergraduate students during the training in the preclinical conservative dentistry should acquire adequate knowledge, skills and attitude which are required for carrying out appropriate activities in dental practice which involves diagnosis treatment and prevention of disease of teeth. During the training program they should be able to identify and use instruments which are used in conservative dentistry and Endodontics. They should also be aware of various restorative procedures with emphasize on tooth conservation.

2. OBJECTIVES

The objectives are dealt under following headings

a. KNOWLEDGE AND UNDERSTANDING:

The student should acquire adequate knowledge during this period of training. Knowledge of the scientific foundation of conservative dentistry and understanding of various treatment procedures carried out in conservative dentistry with emphasize on biological principal to be followed during these treatment procedures and to acquire knowledge of various instruments and materials used in restorative procedures .They should also be aware of various manipulative techniques of restorative material.

b. SKILLS:

The students should be able to demonstrate the following skills which are necessary for practice in conservative dentistry To develop skills in manipulation of various materials used in conservative dentistry. To develop skills in preparation of various cavities and to perform various restorative procedures.

c. ATTITUDE:

The student should be able to apply the current knowledge of various materials used in dentistry in the interest of patients and the community in general. To be aware of recent developments in instruments and materials used in conservative dentistry and update his/her knowledge by attaining various continuing education programmes. Should be aware of both

benefits and health hazards of various restorative materials used in conservative dentistry. Should maintain high standard of professional ethics and apply those in all aspects of professional life.

d. INTEGRATION:

The dental student must be able to identify the healthy and diseased state of the teeth, thereby enabling them to better understand the diseased state and to plan an ideal treatment protocol for the same.

e. KNOWLEDGE ABOUT INFECTION AND CROSS INFECTION IN DENTISTRY

Knowledge about asepsis – disinfection and sterilization of instruments, clinical area / personal care as per universal protection, and disposal of medical wastes in the appropriate modes. Students should be aware of the rules and regulations pertaining to maintenance of clinical set up and waste disposal.

f. COMPUTER PROFICIENCY

Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes. Basic operative skills in analysis of data and knowledge of multimedia. Students should utilize a combination of traditional classroom courses, and online courses. The following validation is required and must be completed.

- i. Technological Requirements for all Graduate Students
- ii. A laptop or desktop computer that supports the following requirements
 - a. Operating system requirements
 - b. Internet browser requirements
 - c. Reliable and consistent access to the internet
 - d. Antivirus software which is current and consistently updated
 - e. Microsoft Office
 - f. Adobe Reader (or equivalent to view PDF files)

3. COMPETENCIES

1. General skills
2. Practice Management
3. Communication and Community Resources

4. Patient Care – Diagnosis
5. Patient Care - Treatment Planning
6. Competencies Specific to the Subject

4. TEACHING HOURS

During IInd year BDS

Lecture	25 hours
Practical	200 hours
Total	225 hours

5. TEACHING METHODOLOGY

Audio Visual Aids: LCD projectors
Identification of instruments used in preclinical dentistry.

Demonstration of various procedures in conservative dentistry.
Demonstration of endodontic procedures in single rooted teeth.

6. THEORY SYLLABUS

TOPIC	MUST KNOW	DESIRABLE TO KNOW	NICE TO KNOW
1.	Introduction to conservative dentistry		
2.	Definition and scope of conservative dentistry and Endodontics		
3.	Classification of cavities		
4.	Nomenclature		
5.	Various chair side positions		
6.	Tooth numbering		

7.	Dental caries		
8.	Restoration -Definition and objectives		
9.	Instrument classification ,nomenclature design formula of hand cutting instrument, grasps and rests		
10.	Rotary cutting instruments, bur design, abrasives and various speeds in rotary instruments. Principle of cavity preparation for (a) Silver amalgam (b)Cast gold inlays (c)Composite resin (d)Glass ionomer		
11.	Matrices, Retainers and wedges		
12.	Separators -Different methods of separation		
13.	Finishing and polishing of restorations		
14.	Management of deep carious lesions- pulp capping and pulpotomy		
15.	Access cavity preparation and brief introduction of instruments used endodontics.		
17.			Infection control
18.			Conservative aesthetic procedures
19.			Bleaching
20.			Complex amalgam restorations
21.			Direct filling gold

Bioethics

Bioethics is the application of ethics to the field of medicine and healthcare. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment, and public health ethics.

7. PRACTICALS:

Practical exercise: 200 hours

Preparation of 1 inch cube in plaster of paris-4 Nos

Preparation of geometric cavities in prepared cubes.

Preparation of tooth models in plaster and preparation of cavities and restoration with modelling wax

a) Incisors -3 Nos

b) Premolars- Upper Premolars -2 Nos; Lower Premolars- 2Nos

c) Molars - Upper Molars 4 Nos; Lower Molars-4Nos

Preparation of Cavities on Extracted Natural Teeth Class I, Class II and MOD and Class V Cavity Preparation. Base Application, Matrix and Wedge Placement, Placement of restoration.

1.	Cavities:		
S no	Cavities	Preparation	Restoration
	Class I	5	5
	Class I with extensions	2	2
	Class II		
	DO conventional	10	10
	MO conventional		
	Conservative preparation in Upper molar		
	Class II MOD	2	2
	Class III and V	4	4 (glass ionomer)
	Class V	2	2(amalgam)

Finishing and polishing of above restorations

Inlay preparation:

Class II preparation

Wax pattern

Sprue attachment

Investment

Casting and finishing

Endodontics

Identification of basic endodontic instruments

Access cavity preparation in upper central incisors

Working length determination

Cleaning and shaping

Obturation of the root canal

Access seal

Demonstration:

Demonstration of class III, class V and incisal edge restoration on extracted teeth with composite resin

Finishing and polishing of the restorations

Identification and manipulation of cavity varnishes, bases like zinc phosphate, zinc poly carboxylate, zinc oxide eugenol cement

Manipulation of glass ionomer cement

Manipulation of amalgam

Identification and demonstration of placement of different types matrix retainers, matrices and tooth separators.

Demonstration of light cure composite and glass ionomer Restoration

Endodontics:

(a) Pulp capping direct indirect on extracted teeth

(b) Pulpotomy on extracted posterior teeth

(c) Root canal access cavity opening on upper Central Incisor (extracted teeth)

Demonstration of instrumentation and obturation of root canal

8. Theory Examination

No Theory Examination

9 .PRACTICAL EXAMINATIONS:

Practical exercise:

Preparation of class II cavity for Silver amalgam in maxillary or mandibular molar tooth (typhodont tooth)

S.no	Excercise	Marks	Time
1	Cavity Preparation	30	45 Minutes
2	Base and Matrix	10	15 Minutes
3	Restoration and Finishing	20	30 Minutes
Total		60 marks	

Viva – voce - 20 Marks

SCHEME OF EXAMINATION:

Internal assessment - 20 marks
Practical - 60 marks
Viva voce - 20 marks
Total - 100 marks

10. FORMATIVE/INTERNAL ASSESSMENT:

The continuing assessment examination held at least 3times in a particular year and best of two examinations shall be considered. The Internal Assessment marks to be submitted to the university, once in every three months. The marks scored by the students shall be displayed on the Notice board and a copy forwarded by HOD shall be sent to the University once in every 3 months.

11. RECORD NOTE / LOG BOOK:

Record shall be maintained and assessed periodically by faculty and HOD. Institution shall provide adequate number of cases/teaching materials as specified in Dental Council of India regulation for the students during clinical / practical training and examinations.

12. TEXT BOOKS:**TEXT BOOKS RECOMMENDED****NAME OF THE BOOKS, AUTHOR, PUBLISHER**

Sturdevant's Art and Science of Operative Dentistry, ELSEVIER

Pre - Clinical Manual of Conservative Dentistry, Dr.V.Gopikrishna, ELSEVIER

8. PRE CLINICAL PROSTHODONTICS & CROWN & BRIDGE

1. GOAL

The dental graduates during training in the institutions should acquire adequate knowledge, necessary skills and reasonable attitudes which are required for carrying out all activities appropriate to general dental practice involving prevention, diagnosis and treatment of anomalies and diseases of the teeth, mouth, jaws and associated tissues. The graduate also should understand the concept of community oral health education and be able to participate in the rural health care delivery programmes existing in the country.

2. OBJECTIVES

a. KNOWLEDGE

- i) Adequate knowledge of the scientific foundations on which dentistry is based and good understanding of various relevant scientific methods, principles of biological functions, ability to evaluate and analyse scientifically various established facts and deals.
- ii) Adequate knowledge of the development, structure and function of the teeth, mouth and jaws and associated tissues both in health and disease and their relationship and effect on general state of health and also bearing on physical and social well being of the patient.
- iii) Adequate knowledge of clinical disciplines and methods which provide a coherent picture of anomalies, lesions and diseases of the teeth, mouth and jaws and preventive diagnostic and therapeutic aspects of dentistry.
- iv) Adequate clinical experience required for the general dental practice.
- v) Adequate knowledge of the constitution, biological functions and behaviour of persons in health and sickness as well as the influence of the natural and social environment on the state of health in so far as it affect dentistry.

b. ATTITUDE

A graduate should develop during the training period the following attitudes.

- i. Willingness to apply the current knowledge of dentistry in the best interest of the patient and community.
- ii. Maintain a high standard of professional ethics and conduct and apply these in all aspects of professional life.
- iii. Seek to improve awareness and provide possible solutions for oral health problems and needs throughout the community.
- iv. Willingness to participate in the CPED programmes to update knowledge and professional skill time to time.
- v. Help and participate in the implementation of the national oral health policy.

c. SKILLS

A graduate should be able to demonstrate the following skills necessary for practice in dentistry.

- i. Diagnose and manage various common dental problems encountered in general dental practice keeping in mind the expectations and the right of the society to receive the best possible treatment available wherever possible.
- ii. Prevent and manage complications if encountered while carrying out various surgical and other procedures.
- iii. Carry out certain investigative procedures and ability to interpret laboratory findings.
- iv. Promote oral health and help prevent oral disease where possible.
- v. Control pain and anxiety among the patients during dental treatment.

d. INTEGRATION

Integrated knowledge about all the divisions in Prosthodontics (CD,RPD,FPD,IMPLANTS etc)

e. KNOWLEDGE ABOUT INFECTION AND CROSS INFECTION IN DENTISTRY

Knowledge about asepsis – disinfection and sterilization of instruments, clinical area / personal care as per universal protection, and disposal of medical wastes in the appropriate modes. Students should be aware of the rules and regulations pertaining to maintenance of clinical set up and waste disposal.

f. COMPUTER PROFICIENCY

Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes. Basic operative skills in analysis of data and knowledge of multimedia. Students should utilize a combination of traditional classroom courses, and online courses. The following validation is required and must be completed.

- i. Technological Requirements for all Graduate Students
- ii. A laptop or desktop computer that supports the following requirements
 - a. Operating system requirements
 - b. Internet browser requirements
 - c. Reliable and consistent access to the internet
 - d. Antivirus software which is current and consistently updated
 - e. Microsoft Office
 - f. Adobe Reader (or equivalent to view PDF files)

3. COMPETENCIES

1. General skills
2. Practice Management
3. Communication and Community Resources
4. Patient Care – Diagnosis
5. Patient Care - Treatment Planning
6. Competencies Specific to the Subject

4. TEACHING HOURS

During Ist Year BDS - 100 Practical hours

During IInd year BDS

Lecture	25 hours
Practical	200 hours
Total	225 hours

5. TEACHING METHODOLOGY

The objectives of teaching microbiology can be achieved by various teaching techniques such as :

- a) Lectures
- b) Lecture Demonstrations
- c) Practical exercises
- d) Audio visual aids
- e) Small group discussions with regular feed back from the students
- f) Integrated Teaching
- g) Symposium and continuing medical education programmes and Computer Aided Study

6. THEORY

I. Introduction to Prosthodontics - Scope and Definition

A. Masticatory apparatus and function:

- 1. Maxillae & Mandible with & without teeth.
- 2. Muscles of mastication and accessory muscles of mastication.
- 3. Brief anatomy of TMJ.
- 4. Mandibular movements.
- 5. Functions of teeth.

B. Various branches of Prosthodontics and prosthesis:

- 1. Scope & limitation.
- 2. Appliances v/s prosthesis.
- 3. Dental prosthesis v/s non-dental prosthesis.

C. Effect of loss of teeth:

- 1. On general health.
- 2. On masticatory apparatus.
- 3. Need of replace lost teeth.

D. Outline of Prosthodontics:

1. Types of Prosthesis.
2. Requirements of prosthesis- Physical, biological, esthetic considerations.

II. Introduction to components of Prosthesis

A. Complete Denture Prosthesis:

1. Various surfaces (Border and surface anatomy).
2. Components - Base and Teeth.

B. Removable Partial Denture:

1. Classification.
2. Major and minor Connectors.
3. Direct retainers.
4. Rests.
5. Indirect retainers.
6. Denture base.
7. Artificial teeth.

C. Fixed Partial Denture:

1. Classification.
2. Retainers.
3. Pontics.
4. Connectors.

III. All related definitions and terminologies from glossary

1. Model
2. Cast
3. Impression
4. Occlusion rim
5. Temporary denture base
6. Permanent denture base
7. Occlusion
8. Face Bow & Articulator
9. Jaw relation - orientation, vertical and centric
10. Christensen's phenomenon
11. Key of occlusion
12. Balanced occlusion
13. Abutment etc...

IV. Introduction to mouth preparation - in brief

A. Complete Dentures

1. General considerations
2. Pre-prosthetic surgery

B. Removable partial dentures

1. General considerations
2. Occlusal rest preparation
3. Modifying contours of the abutments
4. Guide planes
5. Elimination of undercuts

C. Fixed Partial Dentures

1. Principles of tooth preparation - in brief
2. Retainers in brief

V. Introduction to all steps involved in fabrication of Prosthesis

Clinical Steps in brief and laboratory steps in detail

A. Impression Making

1. Definition and requirements and types of impressions
2. Various materials used for different impressions
3. Different theories of impression making

B. Impression Trays

1. Definition, classification, materials, advantages and disadvantages
2. Selection of trays
3. Special trays
4. Spacer design

C. Introduction to jaw relation record

1. Definition and type
2. Temporary denture base - Indications, Advantages, Disadvantages, materials used
3. Occlusion rims - materials, shape, dimensions
4. Clinical procedures of jaw relation recording in brief

D. Articulators and Face bow

1. Basic out line
2. Need for articulators
3. Definition, classification, parts, advantages, disadvantages of articulators
4. Definitions, classification, parts, advantages, disadvantages and purpose of face bow transfer
5. Demonstration of face bow transfer to an articulator on a dummy

E. Selection of Teeth

1. Various guidelines for selection of teeth including dentogenic concept
2. Arrangement of teeth in detail with various factors of esthetics, overjet, overbite etc

F. Occlusion

1. Balanced Occlusion - need and advantages
2. Various factors of balanced occlusion

G. Try in Procedures

1. Anterior try - in
2. Posterior try - in
3. Waxing, carving, polishing and final try - in

H. Processing Procedures

1. Flasking
2. Dewaxing
3. Packing
4. Curing
5. Finishing and polishing of acrylic dentures

VI.Casting Procedures

1. Preparation of die
2. Wax pattern
3. Investing
4. Burnout
5. Casting
6. Finishing and polishing

Bioethics

Bioethics is the application of ethics to the field of medicine and healthcare. Bioethics includes medical ethics, which focuses on issues in health care; research ethics, which focuses issues in the conduct of research; environmental ethics, which focuses on issues pertaining to the relationship between human activities and the environment, and public health ethics.

7. PRACTICAL EXERCISES

1. Preparation of special trays
2. Preparation of temporary and permanent denture bases
3. Preparation of occlusion rims
4. Orientation of occlusion rims on articulator
5. Arrangement of teeth
6. Processing of complete dentures

1. Arrangement of teeth - Must Know
2. Surveying of partially edentulous models and preparing modified master cast - Desirable to Know
3. Preparing of wax patterns spruing, casting and finishing (in batches of students not more than 8)
- Desirable to Know
4. Preparation of plaster models of various preparation of teeth to receive retainers for FPD
- Desirable to Know

5. Prepare wax patterns for minimum of 3 unit FPD and investing, casting and porcelain facing (for Batch of 8 students) - Desirable to Know

Note:

1. Students shall submit one processed denture mounted on an articulator to present on university practical exam along with record book.
2. Exercises of RPD and FPD to be submitted in groups along with the record book

8. Theory Examination

No Theory Examination

9. Practical Examination:

A. Practical Exercise: (Duration-3 hrs) : 60 Marks

Arrangement of teeth in class I relation, Waxing, Carving, Polishing

B. Viva-Voce 20 Marks

C. Internal Assessment 20 Marks

10. FORMATIVE/INTERNAL ASSESSMENT:

The continuing assessment examination held at least 3times in a particular year and best of two examinations shall be considered. The Internal Assessment marks to be submitted to the university, once in every three months. The marks scored by the students shall be displayed on the Notice board and a copy forwarded by HOD shall be sent to the University once in every 3 months.

11. RECORD / LOG BOOK:

Record shall be maintained and assessed periodically by faculty and HOD. Institution shall provide adequate teaching number of cases/teaching materials as specified in Dental Council of India regulation for the students during clinical/practical training and examinations.

12. TEXT BOOKS

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|--|--------------------|
| 1. Essential of Complete Denture Prosthodontics | - Winkler |
| 2. Prosthodontic Treatment for Edentulous Patients | - Zarb Bolender |
| 3. Clinical Removable Partial Denture | - Stewart |
| 4. Fundamentals of Fixed Prosthodontics | - Shillingburg |
| 5. Text Book of Prosthodontics | - Deepak Nallaswam |

13. REFERENCE BOOKS

- | | |
|---|-----------------|
| 1. Impression Techniques for Complete Denture | - Bernard Levin |
| 2. Removable Partial Prosthodontics | - Mc Cracken |
| 3. Contemporary Fixed Partial Denture | - Rosenstiel |
| 4. Syllabus of Complete denture by – Charles M. Heartwell Jr. and Arthur O. Rahn. | |
| 5. Boucher's "Prosthodontic treatment for edentulous patients" | |
| 6. Essentials of complete denture prosthodontics by – Sheldon Winkler | |
| 7. Maxillofacial prosthetics by – William R. Laney | |
| 8. McCracken's Removable partial prosthodontics | |
| 9. Removable partial Prosthodontics by – Ernest L. Miller and Joseph E. Grasso. | |